

CS4234: Optimization Algorithms

@Wxy2003-xy

Contents

1	MAX-CFSU	2
1.1	Approximation factor	2
2	PBS	3
3	ECS	3
3.1	Integer linear programming formulation	3
3.2	Deterministic rounding	4
3.3	Approximation factor proof	4

1 MAX-CFSU

Input: Gateway set J , importance set P , $q \in [0, 1]$

Output: Transmitting gateway set $J^+ \subseteq J$

$J^+ \leftarrow \emptyset;$

foreach $j \in J$ **do**

 At probability q ,
 $J^+ \leftarrow J^+ \cup \{j\};$

return J^+

1.1 Approximation factor

Proof. Define indicator binary variable τ as

$$\tau(j) = \begin{cases} 1, & j \text{ successfully decode} \\ 0, & \text{otherwise} \end{cases}$$

For each gateway, the probability of successful transmission can be expressed as

$$P(j) = q \cdot (1 - q)^{k-1} \cdot k$$

Expectation of the importance from gateway j as

$$\mathbb{E}[\text{importance}(\tau(j))] = p_j \cdot kq(1 - q)^{k-1}$$

and the total importance of all successfully transmitted gateway as

$$\mathbb{E}[\text{importance}(\tau)] = \mathbb{E}[\sum_j \text{importance}(\tau(j))] = \sum_j p_j \cdot kq(1 - q)^{k-1}$$

As we can define q arbitrarily over $[0, 1]$, we will maximize the total expectation by finding the optimal q . Notice that $f(q) = kq(1 - q)^{k-1}$ is concave for $k \geq 2$,

$$\begin{aligned} f'(q) &= k[(1 - q)^{k-1} - q(k - 1)(1 - q)^{k-2}] \\ &= k(1 - kq)(1 - q)^{k-2} \end{aligned}$$

When $f'(q) = 0$, $q = 1$ or $q = \frac{1}{k}$. substitute q with $\frac{1}{k}$,

$$\begin{aligned} f(q)_{\max} &= (1 - \frac{1}{k})^{k-1} \\ (1 - \frac{1}{k})^{k-1} &\geq (1 - \frac{1}{k})^k \geq \frac{1}{e} \end{aligned}$$

We have

$$\mathbb{E}[\text{importance}(\tau)] \geq \frac{1}{e} \sum_j p_j \geq \frac{1}{e} OPT$$

□

Thus the algorithm is

Input: Gateway set J , importance set P ,

Output: Transmitting gateway set $J^+ \subseteq J$

$k \leftarrow |J[0]|$ $J^+ \leftarrow \emptyset;$

foreach $j \in J$ **do**

 At probability $\frac{1}{k}$,
 $J^+ \leftarrow J^+ \cup \{j\};$

return J^+

2 PBS

Define marginal value of adding a new route i to the collection of routes A

$$\Delta(i \mid A) = F(A \cup \{i\}) - F(A)$$

and the value density per cost of route i when being added to A as

$$\delta(i \mid A) = \frac{\Delta(i \mid A)}{c_i}$$

Input: Species set $U = \{u_1, u_2, \dots\}$, route set $\mathcal{S}$, budget B

Output: Total importance

Stack $A \leftarrow \emptyset$;

while $B \geq 0 \wedge \mathcal{S} \neq \emptyset$ **do**

$i_{\max} = \arg \max_{i \in \mathcal{S}} \frac{\Delta(i \mid A)}{c_i}$;
 $A.\text{push}(i_{\max})$;
 $B \leftarrow B - c_{i_{\max}}$;
 $\mathcal{S} \leftarrow \mathcal{S} \setminus \{i_{\max}\}$;

if $B < 0$ *# this guarantees feasibility* **then**

$i' \leftarrow A.\text{top}()$;
 $A.\text{pop}()$;
return $\arg \max_{x \in \{A, \{i'\}\}} (F(x))$;

return $F(A)$;

Approximation factor proof

Suppose the greedily chosen sequence of routes to be

$$i_1, i_2, \dots, i_{t-1}, i_t$$

Given the budget constraint $A = \{i_j \mid j \in [1, t-1]\}$ will have

$$\sum_{i \in A} c_i \leq B < \left(\sum_{i \in A} c_i \right) + c_t$$

as i_t will be the route that causes the budget overflow.

as there can be overlaps between A and any i to be added,

$$F(A \cup \{i\}) \leq F(A) + F(\{i\})$$

3 ECS

Goal

Minimize

$$\sum_{i \text{ activated}} c_i + \sum_{j \text{ not reached}} p_j$$

3.1 Integer linear programming formulation

Define relations $S(i)$ as site i is activated; $R(j)$ as community j is not reached.

Define selection indicator variable x_i as

$$x_i = \begin{cases} 1, & S(i) \\ 0, & \neg S(i) \end{cases} \quad y_j = \begin{cases} 1, & R(j) \\ 0, & \neg R(j) \end{cases}$$

For every j ,

$$\begin{aligned}
(\forall i \in N(j) \neg S(i)) \rightarrow R(j) &\equiv \bigwedge_{i \in N(j)} \neg S(i) \rightarrow R(j) \\
&\equiv \sum_{i \in N(j)} (1 - x_i) - y_j \leq |N(j)| - 1 \\
&\equiv |N(j)| - \sum_{i \in N(j)} x_i - y_j \leq |N(j)| - 1 \\
&\equiv \sum_{i \in N(j)} x_i + y_j \geq 1
\end{aligned}$$

Integer constraint: for every x_i, y_j :

$$x_i \in \{0, 1\} \quad y_j \in \{0, 1\}$$

Goal

Minimize

$$\sum_i c_i x_i + \sum_j p_j y_j$$

3.2 Deterministic rounding

Let the optimal solution of the relaxed LP be x_i^* for i , y_j^* for j , Define a rounding threshold $k \in [0, 1]$:

$$x_i = \begin{cases} 1, & x_i^* \geq k \\ 0, & \text{otherwise} \end{cases}$$

This rounding is clearly polynomial, linear to the size of site set.

3.3 Approximation factor proof

Using this deterministic rounding, in our result $\forall i, S(i) \rightarrow$

$$\begin{aligned}
&x_i^* \geq k \\
&\frac{x_i^*}{k} \geq 1 \\
&\frac{1}{k} c_i x_i^* \geq c_i x_i \quad (x_i \leq 1) \\
\implies &\frac{1}{k} \sum_i c_i x_i^* \geq \sum_i c_i x_i
\end{aligned}$$

For each j that is unreachable, we need to have

$$\begin{aligned}
\forall i \in N(j) (\neg S(i)) &\implies \forall i \in N(j) (x_i^* < k) \\
&\implies \sum_{i \in N(j)} x_i^* < k |N(j)| \leq f k
\end{aligned}$$

From constraint,

$$\begin{aligned}
y_j^* &\geq 1 - \sum_{i \in N(j)} x_i \\
y_j^* &> 1 - fk \\
\frac{1}{1-fk} \cdot y_j^* &> 1 \\
\frac{1}{1-fk} \cdot p_j y_j^* &> p_j y_j \quad (y_j \leq 1) \\
\frac{1}{1-fk} \sum_j p_j y_j^* &> \sum_j p_j y_j
\end{aligned}$$

Combining 2 parts, we have

$$\begin{aligned}
LP &= \sum_i c_i x_i^* + \sum_j p_j y_j^* \\
ALG &= \sum_i c_i x_i + \sum_j p_j y_j \leq \frac{1}{k} \sum_i c_i x_i^* + \frac{1}{1-fk} \sum_j p_j y_j^* \leq \max \left\{ \frac{1}{k}, \frac{1}{1-fk} \right\} LP
\end{aligned}$$

($ALG \leq LP$ as the strict $<$ unreachable community term may be 0)

Find k to minimize $\max \left\{ \frac{1}{k}, \frac{1}{1-fk} \right\}$

Notice that $g_1(k) = \frac{1}{k}$ is strictly decreasing, $g_2(k) = \frac{1}{1-fk}$ is strictly increasing over $k \in (0, 1]$ ($k = 0$ case is trivially impossible), for $f \geq 1$, $\max \left\{ \frac{1}{k}, \frac{1}{1-fk} \right\}$ occurs at the intersection when

$$\begin{aligned}
\frac{1}{k} &= \frac{1}{1-fk} \\
k &= \frac{1}{f+1}
\end{aligned}$$

Given

$$\begin{aligned}
ALG &\leq \max \left\{ \frac{1}{k}, \frac{1}{1-fk} \right\} LP \\
ALG &\leq (f+1) \cdot LP \leq (f+1) \cdot OPT
\end{aligned}$$